

Tapas Ranjan Rath

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EDUCATION

Year	Degree/Certificate	Institute	CPI/%
2019-Present	Ph.D. Cognitive Science	Indian Institute of Technology, Kanpur	7.52/10
2008-2012	B.Tech/Electronics & Instrumentation Engg.	SRM University, Chennai	7.77/10
2007	XII (Higher Secondary Examination (HSE), Odisha)	BJB Junior College, Bhubaneswar, Odisha	75.44%
2005	X (Board of Secondary Education, Odisha)	Project High School, Rengali Dam Site, Angul	90.0%

WORK EXPERIENCE

- **DPS Kalyanpur, Kanpur** (CBSE Physics Teacher) (May'16–June'17)
 - Taught Physics to classes 9-12, designed tests, and demonstrated lab experiments.
- **Chaitanya's Academy, Pune** (Instructor, Physics - IITJEE & NEET) (Aug'14–April'16)
 - Prepared students for IITJEE & NEET, designed test papers.
- **Accenture, Pune** (Senior Software Engineer) (Aug'12–Aug'14)
 - Performed end-to-end manual testing for the British Telecom (BT) project.

RESEARCH EXPERIENCE

- **PhD Research** (Indian Institute of Technology, Kanpur) (2019–Present)
 - Supervisor: Prof. Nisheeth Srivastava, Prof. Narayanan Srinivasan.
 - Thesis: “Investigating the Boundary Conditions of Context Effects in Decision-Making”
 - Developed computational models and Bayesian hierarchical analyses to evaluate decision strategies.

JOURNALS AND CONFERENCES

- **Pair-wise Comparison Difficulty Mediates the Attraction Effect**, T. R. Rath, N. Srivastava, and N. Srinivasan. *Psychonomic Bulletin & Review*. Status: Under Review.
- **Re-evaluating the Numerical-Perceptual Distinction in the Attraction Effect**, T. R. Rath and Vijay Marupudi. *CogSci2025*. Status: Under Review.
- **De-obfuscating Context Effects in Decision-Making: A Simulation Study**, T. R. Rath, N. Srivastava, and N. Srinivasan. Presented at Virtual MathPsych/ICCM, June 2024. mathpsych.org/presentation/1357.
- **Attribute trade-off difficulty modulates the asymmetric-dominance of the decoy**, T. R. Rath, N. Srivastava, and N. Srinivasan. Presented at Annual Conference of Cognitive Science, IIT Bombay, December 2024.
- **Perceptual stimuli with difficult-to-trade-off attribute values show a positive attraction effect**. T. R. Rath, N. Srivastava, and N. Srinivasan. Presented at Annual Meeting of the Society for Judgment and Decision Making, November 22-25, 2024, New York City, USA.

RESEARCH INTERESTS

- Decision-making biases and context effects.
 - Computational modeling of cognitive processes.
 - User sense of agency in recommendation systems.
- Human-AI interactions.
 - Sense of agency while working with agentic AI.
 - Investigating player sense of agency in multiplayer online games.

RELEVANT COURSES AND TECHNICAL SKILLS

Coursework: Neurobiology (BSE656A) Basic Statistics Data Analysis & Inference (CGS602A) Computational Tools for Cognitive Science (CGS600A) Fundamentals of Cognitive Psychology (CGS601A)	Technical Skills: <i>Statistical Analysis:</i> - R (lme4, brms, BayesFactor, ggplot2) - Python (Pandas, NumPy, SciPy, Statsmodels, Seaborn, Matplotlib) <i>Experimental Design:</i> OpenSesame, PsychoPy, JavaScript <i>Eye Tracking:</i> Eyelink 1000 Plus (SR Research) <i>Typesetting:</i> LaTeX
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PROFESSIONAL MEMBERSHIPS

Society for Judgment and Decision Making, Society for Mathematical Psychology, Psychonomic Society, Cognitive Science Society.

POSITIONS OF RESPONSIBILITY

- **Teaching Assistant:** Human-centered Computing (CGS616) (Jan'24 - July'24)
- **Teaching Assistant:** Introduction to Cognitive science (CGS401) (July'23 - Dec'23)